

Lab 1: VibeQueue Product Description

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1 Introduction

1.1 Ecosystem Context

VibeQueue operates within the hospitality industry, specifically targeting small cafes, coffee shops, and local restaurants where atmosphere plays a central role in shaping the customer experience. Background music genre and tempo are among the most controllable environmental factors in a food-service setting, and prior research has linked music choices directly to how long customers stay and how much they spend during a visit (Milliman, 1982; North et al., 1999). Because atmosphere so strongly influences customer satisfaction, dwell time, and per-visit spending, the ability to manage a venue's acoustic environment in real time represents a meaningful, if currently underserved, business opportunity.

1.2 Problem Scope

Over 1 million restaurant and foodservice locations currently operate in the United States, and roughly 9 in 10 of them are small businesses with fewer than 50 employees (National Restaurant Association [NRA], 2023). Because these venues are small, owner-operated businesses, few can absorb a significant new technology investment on top of existing operating costs, and many still rely on whatever ad hoc arrangement keeps customers satisfied in the moment. In practice, staff at these venues are commonly pulled away from other tasks to field song requests in person, which reduces service efficiency and leaves the venue's intended atmosphere dependent on whichever request a staff member happens to honor.

1.3 Current Solutions and Drawbacks

Existing commercial options fail to close this gap. TouchTunes installs its jukeboxes at no upfront cost to a qualifying venue, but the jukebox remains a piece of dedicated hardware that

is owned and serviced by a local amusement operator rather than software the venue controls directly, and eligibility is generally limited to bars, breweries, restaurants, and similar entertainment venues (TouchTunes, n.d.). That operator-dependent, hardware-based model does not give a small, counter-service venue a self-managed digital option. Rockbot offers a subscription-based alternative, with plans starting near \$25 per month per location, but it is built primarily around music that an owner or manager schedules and curates in advance rather than songs customers request directly, so it does not remove staff from the request process the way a customer-facing queue would (Rockbot, n.d.). Many small venues instead rely on manual staff requests, which interrupt employees repeatedly during peak hours and produce inconsistent results whenever staff choose to honor an off-brand request. Others turn to generic personal streaming accounts; however, using a personal subscription such as Spotify Free or Spotify Personal for commercial playback violates the platform's terms of use and exposes the business to legal risk (Spotify, 2023). None of these options gives customers a legitimate, staff-free way to influence the music playing around them.

1.4 Market Gap

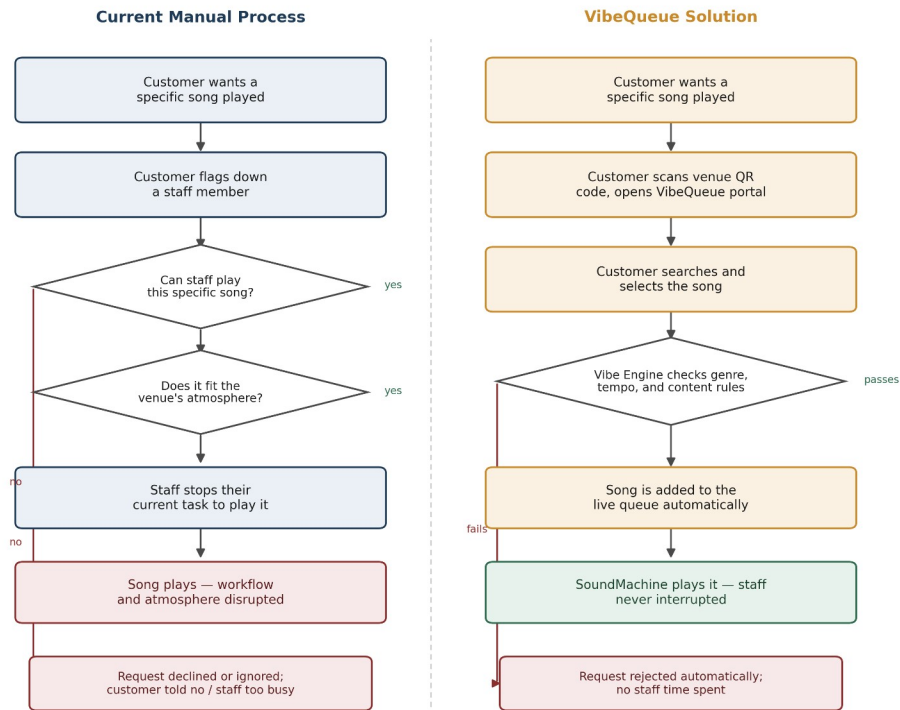
No existing option combines real-time customer input, affordability for a small venue, freedom from staff mediation, and full commercial licensing compliance. Customers have no digital channel through which to influence a venue's atmosphere, and owners have no way to capture a secondary revenue stream from music engagement. Together, these shortcomings leave the small, owner-operated venues that make up the large majority of the industry without a solution built for their scale (NRA, 2023).

1.5 Requirements a Solution Must Address

A solution capable of closing this gap must satisfy five characteristics. First, it must be low-cost enough to be viable on a small business budget, ideally under \$50 per month. Second, it must require no proprietary hardware. Third, it must allow customer-facing interaction without interrupting staff. Fourth, it must comply with commercial music licensing requirements from performing rights organizations such as ASCAP, BMI, and SESAC. Fifth, it must preserve the owner's and staff's ultimate control over the final song selection so that the venue's intended atmosphere remains protected.

1.6 Proposed Solution: VibeQueue

VibeQueue is introduced as this solution: a web-based, hardware-agnostic, subscription jukebox platform for local spaces, built around the guiding principle of “Your Vibes, Managed.” By replacing manual staff requests and unlicensed streaming with a legally compliant, customer-facing digital queue, VibeQueue closes the gap between what small hospitality venues need and what the current market provides. Figure 1 contrasts the manual process most small venues rely on today with the automated flow VibeQueue proposes in its place; the sections that follow describe VibeQueue's product design, its major components, and a representative case study demonstrating how the platform would function in a real venue.

Figure 1*Current Manual Process Flow vs. Solution Flow*

2 Product Description

2.1 Overview

VibeQueue is a software-only digital music ecosystem designed to replace expensive proprietary hardware and manual staff processes for small hospitality venues. Rather than requiring a venue to purchase or lease dedicated equipment, the platform is deployable entirely on devices the business already owns, replacing ad hoc, staff-mediated song requests with a seamless, customer-facing digital queue. Section 1.6 introduced VibeQueue by name; this section describes the product in greater depth, beginning with its goals, its differentiation from existing options, and its specific features and components.

2.2 Goals and Objectives

The platform pursues five core objectives: giving customers a frictionless way to influence a venue's music through their own smartphones; giving owners and staff complete control over which songs can be requested; providing an optional secondary revenue stream through a pay-to-request model; ensuring that all music playback is legally licensed for commercial use; and remaining deployable with zero new hardware purchase. By utilizing a subscription-based pricing model and running entirely on devices the business already owns, VibeQueue offers small venues a low-cost, legally compliant alternative to traditional jukeboxes and unlicensed personal streaming.

2.3 Key Differentiators vs. Competition

VibeQueue differentiates itself from existing options along four dimensions. First, because the platform is software-only, it runs entirely on devices the venue already owns rather than depending on a dedicated jukebox unit installed and serviced by an outside operator, unlike

TouchTunes (TouchTunes, n.d.). Second, VibeQueue is built around direct customer song requests rather than owner-scheduled playlists, unlike Rockbot's manager-curated model (Rockbot, n.d.). Third, VibeQueue puts the request itself directly in the customer's hands through the self-service interface detailed in Section 2.4, replacing the verbal, staff-mediated requests that define the manual-request status quo. Fourth, VibeQueue builds commercial licensing into the platform from the start, as described in Section 2.5, in contrast to venues that rely on unlicensed personal streaming accounts (Spotify, 2023). Taken together, these four dimensions position VibeQueue as the only option that is simultaneously hardware-free, customer-driven, staff-free, and fully licensed for commercial playback.

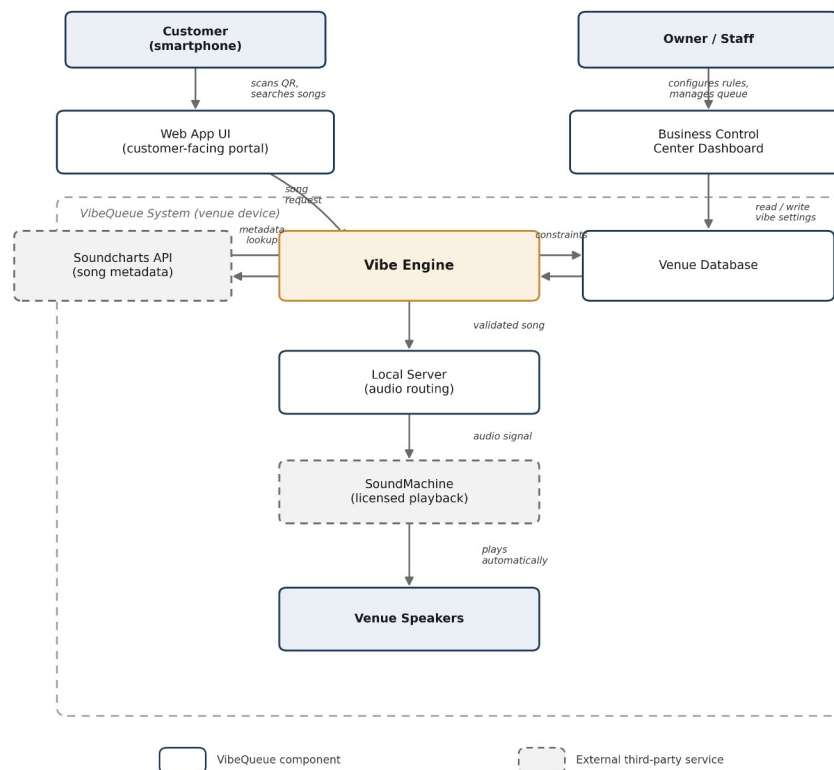
2.4 Key Product Features and Capabilities

Key features include QR Code Access, which allows customers to open the venue's specific VibeQueue portal in a mobile browser without downloading a dedicated application. Customers can browse and request songs using metadata from the Soundcharts API, with requests instantly filtered by VibeQueue's proprietary Vibe Engine. This engine evaluates songs against the venue's allowed genres, tempos, moods, and explicit content settings, automatically rejecting non-matching requests before they enter the queue. Owners manage these settings through the Business Control Center dashboard, which also features Dynamic Vibe Moderation to enforce content rules system-wide, along with queue management, user-session controls, and analytics on the most requested songs and peak request times. Additionally, the platform includes an optional Pay-to-Request feature, allowing customers to pay a small transaction fee, typically \$0.99 to \$1.99, to boost their song to the top of the queue, with the business retaining the majority of the revenue and VibeQueue collecting a modest platform fee to sustain development. Every song that reaches the queue is delivered through SoundMachine's licensed

streaming service, so venues never have to manage commercial licensing on their own. What VibeQueue does not do is give customers final authority over playback: every request must first satisfy the Vibe Engine and the owner's configured constraints before it is added to the queue, ensuring the venue's intended atmosphere is preserved even as customers participate.

2.5 Major Components (Hardware/Software)

VibeQueue requires no proprietary hardware; it runs on any internet-connected device the venue already owns, such as a tablet, smartphone, or laptop, connected to the venue's existing speaker system. Figure 2 presents the Major Functional Component Diagram (MFCD), which maps VibeQueue's software architecture and traces how data moves between components, distinguishing the components VibeQueue itself builds from the external services it depends on. At the center of that architecture are a Web App UI for the customer-facing portal and the Vibe Engine for song validation. The Vibe Engine queries the Soundcharts API for commercially safe music metadata, including genre, tempo, mood, popularity, and explicit-content flags (Soundcharts, n.d.), and checks that metadata against the Venue Database, which stores each venue's vibe settings, song-queue state, and transaction and revenue data. A Local Server handles audio routing and communication between the Vibe Engine and SoundMachine, the third-party service that performs licensed commercial audio playback (SoundMachine, n.d.). The Business Control Center Dashboard gives owners and staff direct read and write access to the Venue Database, so constraint changes made on the dashboard take effect on the next customer request.

Figure 2*VibeQueue Major Functional Component Diagram (MFCD)*

As Figure 2 shows, two paths feed into the Vibe Engine at the center of the diagram. On the customer side, a customer's smartphone reaches the Web App UI, which forwards each song request to the Vibe Engine; on the owner side, the Business Control Center Dashboard reads and writes the venue's vibe settings directly in the Venue Database. The Vibe Engine reconciles these two paths by checking every incoming request's metadata, retrieved from the external Soundcharts API, against the constraints stored in the Venue Database. A request that passes is forwarded as a validated song to the Local Server, which routes it to SoundMachine, the external licensed-playback service, for automatic delivery to the venue's speakers; a request that fails is rejected before it ever reaches the queue. The dashed borders around Soundcharts and

SoundMachine in Figure 2 mark them as third-party services VibeQueue depends on but does not own, while the solid borders identify the components VibeQueue itself builds and operates.

2.6 Technology Stack

The entire system is built on a LAMP technology stack, using Linux as the operating system, Apache as the web server, MySQL as the underlying database platform, and PHP as the primary scripting language. Development tools include a QR code generation library for venue-specific portal linking and a mobile-first, responsive web front end. The initial prototype explored the Spotify API for song metadata but pivoted to Soundcharts and SoundMachine after determining that Spotify's commercial-use licensing restrictions were incompatible with VibeQueue's business model (Spotify, 2023).

3 Identification of Case Study

3.1 Case Study Participants

The primary case study for VibeQueue is “Brewpoint Café,” an independent coffee shop with approximately 30 seats that primarily serves morning commuters and remote workers between 7 a.m. and 2 p.m. Three roles participate in the case study. Jamie, the café's owner, configures the venue's genre and tempo constraints through the Business Control Center and reviews queue activity between rushes. Two baristas staff the counter during the morning rush and, before VibeQueue, were the ones fielding verbal song requests directly. Alex, a regular customer, represents the café's typical patron: someone who wants a say in the music without pulling a barista away from the register.

3.2 Problem Being Addressed

Prior to adopting VibeQueue, Brewpoint's baristas were pulled away from other tasks several times during a typical morning rush by verbal song requests, and Jamie had no reliable way to preserve the café's intended acoustic, lo-fi atmosphere whenever staff accommodated an off-brand request. The café also had no way to turn customer interest in the music into any kind of revenue, since a verbal request costs the business time without generating anything in return.

3.3 Features Exercised

The case study exercises five core features: QR Code Access from table cards at each seat, Song Browsing and Requests powered by the Soundcharts API, the Business Control Center configured with a genre allowlist limited to acoustic, indie, and lo-fi tracks with explicit content blocked and a maximum tempo of 120 beats per minute, Dynamic Vibe Moderation to enforce those rules automatically, and the optional Pay-to-Request boost priced at \$0.99. For

Alex, these features yield a direct, self-service way to influence the room's music. For the baristas, they yield zero interruptions during the morning rush. For Jamie, they yield both a protected brand atmosphere and a new, if modest, revenue stream from the Pay-to-Request fee.

3.4 Use Case Scenario

In the scenario, Alex sits down, notices the QR card on the table, and scans it with a personal smartphone. The VibeQueue portal opens directly in the browser with no download required, displaying the current queue and a search bar. Alex searches for “Golden Hour” by JVKE; the Vibe Engine checks the request against Brewpoint's constraints, confirming that the genre (pop/indie) is allowed, the track is not explicit, and the tempo of 98 beats per minute falls under the 120-beat-per-minute limit, after which the song is added to the queue. Alex optionally pays \$0.99 to boost the song to the top of the queue, with payment processed automatically. Staff experience no interruption, and the song plays through SoundMachine directly on the café's speakers. During a slower period, Jamie logs into the Business Control Center on the counter tablet, reviews the queue and request analytics, and adjusts the genre allowlist in preparation for the afternoon shift.

3.5 Future Expansion and Broader Customer Base

Beyond Brewpoint Café, VibeQueue is intended to serve a broader customer base of small, atmosphere-conscious venues, including small café chains, college dining halls, hotel lobbies, fitness studios, and barbershops. A full-capability version of the platform could extend the Business Control Center into a cross-location analytics view, letting a chain owner compare request volume, top genres, and Pay-to-Request revenue across every location from a single dashboard rather than configuring each café separately. Longer-term opportunities include a B2B

SaaS licensing model for venue management companies and loyalty features that reward customers with “vibe points” for queued requests.

Glossary

API (Application Programming Interface): A set of rules and protocols that allows different software applications to communicate with each other.

ASCAP (American Society of Composers, Authors, and Publishers): A U.S. performing rights organization that licenses music for commercial use.

BMI (Broadcast Music Inc.): A U.S. performing rights organization that manages commercial music licensing on behalf of songwriters and publishers.

Business Control Center: VibeQueue's owner-facing web dashboard for configuring vibe constraints, managing the song queue, and viewing analytics.

Hardware-Agnostic: Software designed to run on any existing device without requiring specific or proprietary hardware.

LAMP: An open-source web development stack comprised of Linux (OS), Apache (web server), MySQL (database), and PHP (scripting language).

MFCD (Major Functional Component Diagram): A visual representation of a system's major components and the data flows between them, as defined in CS 410.

QR Code (Quick Response Code): A type of two-dimensional matrix barcode that can be scanned by a smartphone camera to open a URL or application.

SESAC (Society of European Stage Authors and Composers): A U.S. performing rights organization providing commercial music licensing.

Soundcharts API: A third-party music analytics API that provides commercially safe song metadata including genre, tempo, mood, popularity, and explicit content flags.

SoundMachine: A third-party commercial music streaming service that holds ASCAP/BMI/SESAC licenses; used by VibeQueue for legal audio playback in venues.

Vibe Engine: VibeQueue's proprietary, in-house song selection and validation algorithm; evaluates customer song requests against venue-configured constraints before allowing them into the queue.

References

- Milliman, R. E. (1982). Using background music to affect the behavior of supermarket shoppers. *Journal of Marketing*, 46(3), 86–101. <https://doi.org/10.2307/1251706>
- National Restaurant Association. (2023). *Restaurant industry facts at a glance*. <https://restaurant.org/research-and-media/research/industry-statistics/national-statistics>
- North, A. C., Hargreaves, D. J., & McKendrick, J. (1999). The influence of in-store music on wine selections. *Journal of Applied Psychology*, 84(2), 271–276. <https://doi.org/10.1037/0021-9010.84.2.271>
- Rockbot. (n.d.). *Business music, TV & digital signage solutions*. <https://rockbot.com>
- Soundcharts. (n.d.). *Music analytics platform*. <https://soundcharts.com>
- SoundMachine. (n.d.). *Licensed background music for business*. <https://sound-machine.com>
- Spotify. (2023). *Spotify terms and conditions of use*. <https://www.spotify.com/us/legal/end-user-agreement/>
- TouchTunes. (n.d.). *Frequently asked questions*. <https://business.touchtunes.com/faq>